The Holloway Gambit: The Willing Sacrifice in Split Conformal Prediction

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Abstract

Conformal prediction has been touted as a more formal, rigorous approach to adding uncertainty to a forecast. The sole objective of this note is to point out that rigor cuts both ways in the case of residual pooling, the technique used in the vast majority of conformal prediction applications. The fact that unconditional guarantee of coverage is provided is not in question, but we make clear, we believe for the first time, that there is an opposing guarantee too: a permanent gambit of logarithmic-score regret which no amount of data or tuning can subsequently reduce. We give the exact size of the sacrifice, and a financial reading of it as the growth rate of an oracle adversary betting against odds set by someone using residual pooling.

1 The claim

In recent years conformal prediction has gained attention in the machine learning community in particular. However, there is a danger this emphasis becomes a substitute for careful statistical modeling. The use of split conformal prediction, where the field started, is essentially synonymous with the statistical advice that says one's model residuals are best modeled by their past empirical distribution. Such advice is arguably naive in fields like finance where stochastic volatility is a mainstay. On the other hand there are many well-used algorithms in classification, to pick one area, where the off the shelf outputs ("proba") tend to be very poorly calibrated, and in those settings adding a conformal wrapper has real utility.

Our point is not to persuade anyone against pragmatism. It is that there can be no free lunch, and that there is nothing sacred about the empirical distribution however practical it may prove as a first remediation. The purpose of this note is to convert that common sense into a crisp mathematical statement.

It should go without saying that the split conformal guarantee is not in question. Given exchangeable data, a fitted predictor, and a nonconformity score, split conformal prediction returns a set $C_\alpha(x)$ with

$$\mathbb{P}\big(Y_{n+1} \in C_\alpha(X_{n+1})\big) \;\geq\; 1-\alpha, \tag{1}$$

finite-sample and with no assumption on the data distribution (Vovk et al., 2005; Lei et al., 2018; Angelopoulos and Bates, 2023). The danger lies solely in the *interpretation* of (1) when a user wants a conditional distributional prediction of the next outcome. The averaging here takes place over both X_{n+1} and Y_{n+1}, and that is not at all the usual notion of prediction. The paper Lei

*Global Strategic Minerals Corporation. Email: `peter.cotton@gsmc.ai`. Code and figures are at https://github.com/microprediction/conformalprediction. A longer treatment, with the omitted readings of the identity and the time-series discussion, is Cotton (2026).

and Wasserman (2014) is one of many warnings about this in the literature, where it is stated that "a good estimator must satisfy something more than marginal coverage".

The claims nonetheless get stronger as one descends from the journals toward the practitioner. One widely used forecasting library tabulates methods by "Calibration Guarantee", awarding the check mark to conformal prediction and a tilde to bootstrap, quantile regression and ARIMA alike (Nixtla, 2026). A popular introduction adds that with Bayesian posteriors or bootstrapping "we have no guarantees that these approaches will perform well on new data" (Molnar, 2023). From there it is a short step to conformal prediction as the thing that "converts any statistical, machine or deep learning model into a probabilistic prediction model" (Manokhin, 2023). That is the step we think cannot be taken for free, and below we price it.

2 Setup

Let $(X_i, Y_i)_{i=1}^{n+1}$ be exchangeable. Fix a point predictor $\hat{\mu}$ trained on a disjoint set, and a *nonconformity score* $s(x, y)$. The canonical choice in regression is the absolute residual $s(x, y) = |y - \hat{\mu}(x)|$. Compute calibration scores $s_i = s(X_i, Y_i)$ for $i = 1, \ldots, n$ and let

$$\hat{q} \;=\; s_{(k)}, \qquad k = \lceil (n+1)(1-\alpha) \rceil, \tag{2}$$

the k-th order statistic (with $\hat{q} = +\infty$ when $k > n$). The split-conformal set is

$$C_\alpha(x) = \{y : \; s(x, y) \le \hat{q}\} \;=\; [\hat{\mu}(x) - \hat{q},\; \hat{\mu}(x) + \hat{q}] \tag{3}$$

for the absolute-residual score. Exchangeability of the augmented scores makes the rank of s_{n+1} uniform, which yields (1) (Vovk et al., 2005; Lei et al., 2018). None of this is in dispute. The question is what it is taken to mean.

The guarantee one can have distribution-free is the marginal one, and a marginally valid band "tends to overestimate the set when x is in the high density area and to underestimate for low density x" (Lei and Wasserman, 2014). It over-covers the easy inputs and under-covers the hard ones. That's the game, and Figure 1 shows it on heteroscedastic data: 90% overall, near-100% where the noise is small, well below target where it is large. Conditional coverage is not a repair one can buy, since a band with non-trivial finite-sample conditional validity has infinite expected length at almost every point of a continuous distribution (Lei and Wasserman, 2014), a result refined by Foygel Barber et al. (2021).

3 Coverage does not constrain the score

Marginal validity is not, on its own, evidence of a good method. We would argue that coverage and log-likelihood belong on separate axes, as the two separate things a forecaster reports. Those can be a point predictor $\hat{\mu}(x)$ and a full predictive density $f(\cdot \mid x)$. The canonical residual score $s(x, y) = |y - \hat{\mu}(x)|$ uses only $\hat{\mu}$, the *location* of the forecast. It never sees the spread or shape of f, so the conformal set and its coverage are invariant to exactly the part of f that the log-score rewards.

Remark 1 (Residual-score coverage does not constrain the log-score)**.** Two forecasters with the *same* location $\hat{\mu}$ but different predictive densities f have *identical* conformal sets, and identical marginal coverage at every level α, while their expected log-scores $\mathbb{E}[\log f(Y \mid X)]$ may differ

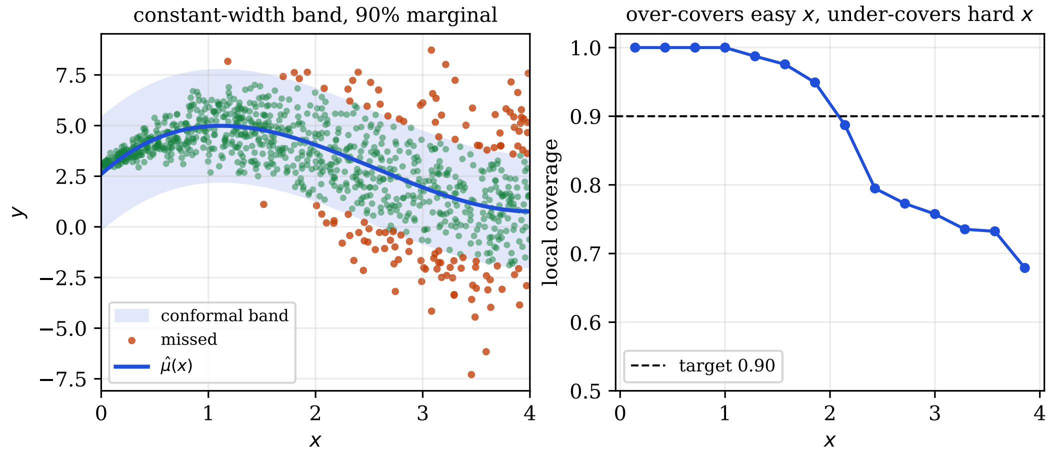

Figure 1: Marginal is not conditional. A constant-width split-conformal band (left) attains 90% *marginal* coverage on heteroscedastic data, but its *local* coverage (right) runs near 100% where the noise is small and well below target where it is large.

by an arbitrary amount. Take X degenerate and $Y \sim N(0,1)$, so both use $s(x,y) = |y|$ and the conformal set is the fixed interval $[-\hat{q}, \hat{q}]$ whatever density is claimed. With $f_\sigma = N(0,\sigma^2)$, $\mathbb{E}[\log f_\sigma(Y)] = -\frac{1}{2}\log(2\pi\sigma^2) - 1/(2\sigma^2)$ diverges to $-\infty$ at both ends of the σ range, while the set never moves.

This is nothing more than a formal version of a point numerous authors have been making informally. Perhaps the most blunt of these commentators is Recht (2024), who writes that conformal validity "almost invites you to use garbage prediction functions". How could it be otherwise? The spread never entered (2).

4 The gap

A conformal predictor can be made to output a distribution rather than a set, the *conformal predictive system* (CPS) of Vovk et al. (2019), built from *signed* residual scores. We view it as an attempt to fix the residuals subject to the restriction of using only one unconditional shared shape, as compared to a data-dependent scale such as one would see in a GARCH model. It is the mode in which conformal prediction estimates a full distribution, so it is the form the criticism must address.

Fix a location predictor $\hat{\mu}$ with residual $R = Y - \hat{\mu}(X)$ of marginal density g and CDF G. In the large-calibration limit the CPS predictive distribution is $\Pi_x(y) = G(y - \hat{\mu}(x))$, with density $\pi_x(y) = g(y - \hat{\mu}(x))$. This is the base location forecast re-leveled by the marginal signed-residual law, and the estimation in it was all done by the base model and the residual sample. Conformal supplies the leveling and nothing else. Write the true conditional residual density as $r(\cdot \mid x)$, so the oracle density is $q^\star(y \mid x) = r(y - \hat{\mu}(x) \mid x)$, and let $\bar{r} = g$ be the marginal residual density.

Proposition 1 (The residual-information gap)**.** *With $I(R;X) = \mathbb{E}_X \, \mathrm{KL}(r(\cdot \mid X) \,\|\, \bar{r}) \geq 0$ the mutual information between the residual and the input, and assuming the relevant densities and information quantities are finite,*

$$\mathbb{E}[\log q^\star(Y \mid X)] - \mathbb{E}[\log \pi_X(Y)] \;=\; I(R;X). \tag{4}$$

Among all single-shape forecasters $y \mapsto h(y - \hat{\mu}(x))$, $\mathbb{E}[\log h(R)]$ is maximized at $h = \bar{r}$, and the signed CPS attains this optimum. Hence no recalibration that ignores X can reduce $I(R; X)$. Reducing it requires conditioning on X.

Proof. Let $\mathcal{R}(h)$ be the expected log-score regret of the single-shape forecaster $y \mapsto h(y - \hat{\mu}(x))$ relative to the oracle. The oracle scores Y by $\log q^\star(Y \mid X) = \log r(R \mid X)$ and this forecaster by $\log h(R)$. The joint $P_{X,R}$ has density $p(x)\, r(\rho \mid x)$ and the product reference $P_X \otimes H$ has density $p(x)\, h(\rho)$, so

$$\mathcal{R}(h) = \mathbb{E}\left[\log \frac{r(R \mid X)}{h(R)}\right] = \mathrm{KL}\big(P_{X,R} \,\|\, P_X \otimes H\big) = I(R; X) + \mathrm{KL}(\bar{r} \,\|\, h), \tag{5}$$

the last step the chain rule for relative entropy along a product reference, the R-marginal of $P_{X,R}$ being $\bar{r}$. Taking $h = \bar{r}$ kills the second term and gives (4). That second term is ≥ 0 and is minimized at $h = \bar{r}$ by Gibbs, so no X-blind shape improves on $I(R; X)$, and $I(R; X) = \mathrm{KL}(P_{X,R} \,\|\, P_X P_R)$ is reduced only by conditioning on X. □

The theorem is a small one. Its interest is in what it prices. The right-hand side of (4) is exactly the information about the residual distribution that remains in X after the location predictor is fixed, and calling it the conformal information gap gives "blind to conditional residual shape" a number. Note what the result does not say. The signed CPS is log-score optimal among single-shape location forecasters, so conformal prediction is not dominated within its own class. What it pays for is the class restriction, not the calibration step. Reading ordinary absolute-residual intervals as a distribution costs an additional $\mathrm{KL}(\bar{r} \,\|\, h_{\mathrm{sym}})$ for discarding sign information, at most one bit; the structural loss is $I(R; X)$.

A false-pooling cost. A single-shape system asserts that, once $\hat{\mu}$ is fixed, one residual law fits everyone. That is the assertion $R \perp X$. The gap is the relative entropy from the true residual experiment to the nearest such independence model, and one pays it whether or not one notices making it.

A betting rent. By the log-optimal (Kelly) growth theorem, a gambler who knows a distribution P and bets against a fair book priced by Q compounds wealth at expected log-rate $\mathrm{KL}(P \,\|\, Q)$ per round (Kelly, 1956; Cover and Thomas, 2006). Read $\bar{r}$ as the book posted by a single-shape conformal forecaster and $r(\cdot \mid X)$ as what the oracle knows. The oracle's expected rent per round is $\mathbb{E}_X\, \mathrm{KL}(r(\cdot \mid X) \,\|\, \bar{r}) = I(R; X)$. The conformal step fixes the level of the book, meaning its normalization and hence its coverage. It does not touch the rent.

A natural worry is that $I(R; X)$ merely measures how badly $\hat{\mu}$ fits. It does not. Take the Bayes-optimal location $\hat{\mu}(x) = \mathbb{E}[Y \mid X = x]$, which zeroes the conditional mean of R. Under heteroscedasticity the *conditional law* of the residual still varies, so $I(R; X) > 0$ in general. The gap measures conditional residual scale and shape, and that is orthogonal to getting the location right.

5 What follows in practice

Coverage error and proper-score performance are two coordinates, not one. Residual split conformal moves a report horizontally, changing the set so that coverage error goes to zero and leaving the density, and hence the score, exactly where it was. Conformal predictive systems do have a vertical coordinate, but it is capped at the oracle minus $I(R; X)$. Fitting shape to a

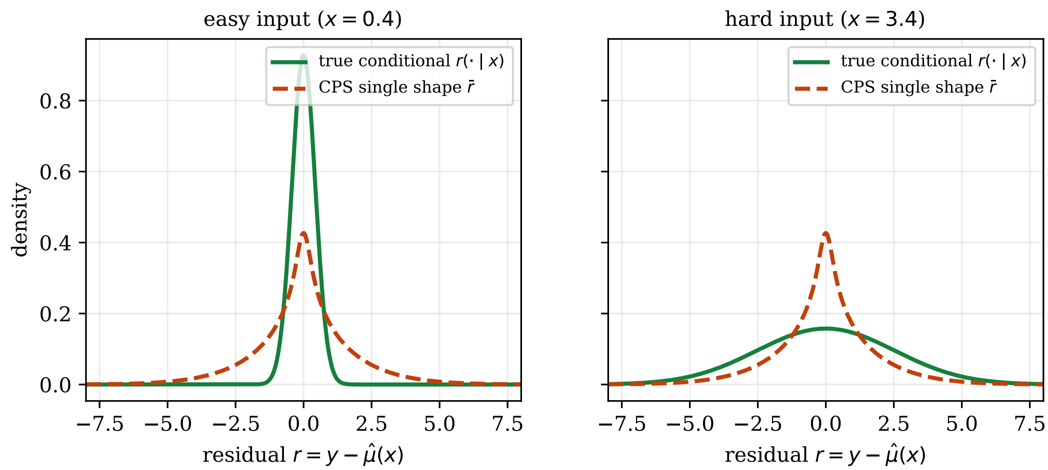

Figure 2: The gap. A single-shape conformal predictive system uses the marginal residual law $\bar{r}$ for every input. The oracle uses the true conditional $r(\cdot \,|\, x)$, narrow for easy inputs (left) and wide for hard ones (right). The expected log-score regret is exactly $I(R; X)$.

proper score moves a report up, and conditioning on x is the only way to raise it past that cap. A system reported by its coverage alone is a system reported by one coordinate of a two-coordinate quality.

Some consequences follow. In the post-hoc setting considered here, conformalize *last*. A conformal step applied to a residual score changes the set and leaves the predictive density, and hence its score, untouched (Remark 1), so a proper score is improved by estimating and never by certifying. The conformal predictive system is the exception that proves the point. It does supply a density, and by Proposition 1 it is the best single shape available, but its score stops at the oracle minus $I(R; X)$.

To close that gap one must condition on X. The cheapest way to do so is inside the conformal step itself, by replacing the plain residual with a normalized score $s(x, y) = |y - \hat{\mu}(x)|/\hat{\sigma}(x)$ (Lei et al., 2018), which is exactly the move that leaves the single-shape class. Whether one then calls the result conformal prediction or heteroscedastic modeling is a matter of labelling. The sharpness came from $\hat{\sigma}$.

Scope. Our strong claims target one object, *post-hoc, marginal, split conformal prediction with a residual score.* Conditional and shape-adaptive constructions leave the single-shape class and our result does not apply to them. Examples include conformalized quantile regression (Romano et al., 2019), Mondrian and binned conformal predictive systems (Boström et al., 2021; Toccaceli, 2026), and conformal training (Stutz et al., 2022). These can buy sharpness by conditioning on X or optimizing a proper objective and trading unconditional guarantees, but so can almost every statistical approach. Where coverage genuinely is the deliverable, as in selective prediction, shortlisting, or conformal p-values for novelty detection (Angelopoulos et al., 2021; Bates et al., 2021), the method is not merely sound but often ideal. Finally, the exact mutual-information form belongs to the log-score; for CRPS or the interval score the analogous gap is the regret of the best X-blind residual law under that score, which is not an information quantity.

6 Conclusion

Conformal prediction certifies the coverage of a set, finite-sample and distribution-free, under exchangeability. That guarantee is genuine and, for the right objective, exact. In this note we go beyond the usual caution (against reading the guarantee as forecast quality) and we have emphasized the permanent structural concession made at the point residual pooling is chosen. Our message is only that nothing comes without a price. The unconditional coverage guarantee is accompanied by an equally rigorous guarantee that the conformal information gap will never be crossed.

Within the single-shape residual class conformal prediction is optimal, and where a guarantee is the product it is the right tool. But coverage answers the question *is the truth in this region?*, and in forecasting that is frequently not the question anyone was asking. This is not weak uncertainty quantification. It is exact uncertainty quantification for a set, a coverage guarantee, no more and no less. The gambit is permanent. Its price is $I(R; X)$, and the modeler should know what has been paid.

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